Value and Quality in Healthcare

MHIA 6160 | 1.50

Dr. Cameron B. Chiarot

Winter 2027

Course Information

Course:	MHIA 6160, 1.5 Credits, 6 Weeks, 3-Hour Sessions
Instructor:	Dr. Cameron B. Chiarot
Email:	TBD
Location:	Schulich School of Business, York University (Keele Campus)
Room:	TBD
Day & Time:	Wednesdays, 6:30 – 9:30 PM
Office Hours:	Wednesdays, 6:00 – 6:30 PM (in the lecture room)
Lecture Format:	In-person, Synchronous
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I Professor Profile & Research Interests

Dr. Cameron B. Chiarot is a distinguished professional in global health economics, epidemiology, and biostatistics, combining rigorous scientific inquiry with implementation science. He integrates extensive methodological expertise with over twenty years of experience managing research capital exceeding CAD$100 million (USD$73 million) and leading large-scale organizational transformations within highly interdisciplinary teams in Canada's private and public healthcare sectors. His research primarily concentrates on Health System Resilience and infectious disease modelling. Dr. Chiarot developed the *econosyndemic* framework to empirically examine how complex macroeconomic and microeconomic conditions interact to worsen health outcomes, inequalities, and inequities. His scientific portfolio investigates infectious diseases such as HIV, malaria, and COVID-19, as well as the maternal care continuum, child health, and outpatient services (i.e., Utilization). Dr. Chiarot possesses executive experience in strategic planning, financial modelling, profit and loss (P&L) management, capital projects, and stakeholder engagement within complex private, public, and academic healthcare organizations. He has extensive international experience, including Canada, the United States, the United Kingdom, and the Republic of Zambia. Dr. Chiarot holds a PhD from the University of Waterloo's School of Public Health Sciences, an MBA with Distinction from York University's Schulich School of Business, an Honors Bachelor of Science from the University of Western Ontario, and a Certificate in Healthcare Executive Leadership from the University of Toronto's Rotman School of Management.

II Course Overview

This course delineates the principal components of a value-based healthcare system. It covers medical condition-focused delivery models, cost analysis and remuneration, as well as Information Technology (IT) infrastructure, systems integration, and the geography of care. This course involves students in understanding and managing these elements and exploring their application within the Canadian healthcare framework. Building on these foundations, the course adopts a *dual-lens* strategy to navigate the complex interface between Canada's public health mandates and private-sector innovations. The curriculum extends beyond standard definitions to examine how value and quality are influenced by the interaction of economic conditions, digital transformation, and health system resilience. Students will engage with the multi-dimensional landscape of contemporary healthcare delivery, acquiring skills to synthesize theoretical frameworks with the practical realities of implementation. Although the course is rooted in health systems theory, it also introduces coding software (i.e., `R`) to facilitate visualization and analysis of the complex value and quality datasets essential for evidence-based leadership.

Course Prerequisites: None.

III Pedagogical Philosophy

This course utilizes a specialized pedagogical approach designed for high-performance leaders in health systems and administration. Rather than focusing on theory in isolation, the curriculum is built upon two foundational pillars:

1. The *Dual-Lens* Strategy (Theory & Practice)
2. The *Code-Along* Philosophy (Evidence-Based & Applied)

The *Dual-Lens* Strategy (Theory & Practice)

To navigate the complex interface between public mandates and private innovations, each module is examined through two distinct perspectives: the **System Lens** (Public Sector), which emphasizes macro-level equity, the Canada Health Act, population-centered care, and regulatory frameworks that safeguard health system resilience; and the **Industry Lens** (Private Sector), which concentrates on micro-level efficiency, technological innovation (e.g., artificial intelligence and digital infrastructure), fiscal sustainability, and the competitive factors driving value.

The *Code-Along* Philosophy (Evidence-Based & Applied)

While the course introduces R and coding software, it is **not** a computer science class. The pedagogical focus is on data literacy for leadership, which is based on standardized scaffolding. In this approach, students are not expected to write complex code from scratch; rather, a series of weekly R-Laboratory sessions utilizes a *Code-Along* model where standardized scripts are provided. The objective is for students to learn how to modify parameters and interpret outputs to drive evidence-based insights, ensuring they can lead technical teams effectively without needing to be full-time programmers.

Simulation-Based Synthesis (DAEDALUS)

The course concludes with a simulation-based capstone utilizing the DAEDALUS modelling interface. This sandbox environment enables students to apply the *dual-lens* strategy to a real-time crisis, evaluating how strategic levers influence both health outcomes and economic value prior to presenting their findings to an executive-level audience.

IV Course Learning Outcomes

Upon successful completion of this course, students will be able to:

1. **Evaluate** how the legal and regulatory pillars of the Canada Health Act influence the delivery of high-quality care within centralized and decentralized frameworks.
2. **Synthesize** the relationship between potential access and realized utilization to identify systemic barriers that diminish healthcare value.
3. **Analyze** how macroeconomic shifts and microeconomic drivers impact the fiscal sustainability and quality of health service delivery.
4. **Critique** innovative delivery models based on their ability to optimize value through the lens of equity and population-centred care.
5. **Develop** resilient risk management strategies aligned with the Sustainable Development Goals (SDGs) to safeguard the long-term quality of health systems and promote environmental sustainability.
6. **Evaluate** the strategic deployment of artificial intelligence (AI) and digital infrastructure to ascertain their effects on clinical quality and operational value from both patient-centred and system-wide perspectives.
7. **Utilize** coding software to visualize and interpret complex health datasets, thereby transforming raw data into evidence-based insights that enhance value and quality.

V Course Materials

1. Please refer to the **weekly reading list**.
2. Additional readings on request.
3. There is **no** assigned textbook for this course.

VI Modules & Curriculum Map

Module 1: Foundations of Value & Quality (Week 1)

1. The Canadian Health Act & Framework
2. Centralized versus Decentralized Governance
3. Public versus Private Sector Interface
4. The Value Framework & Defining Value (Quantitatively)

Module 2: Measuring Value & Quality (Week 2)

1. Access versus Realized Access (Utilization)
2. Economic Drivers of Utilization (Macro- and Micro-Conditions)
3. Evaluating and Measuring Utilization
4. The Effectiveness-Efficiency Frontier

Module 3: Population Health & Delivery Innovations (Week 3)

1. Equity versus Equality Frameworks and Empirical Measurement
2. Patient-Centred versus Population-Centred Care
3. Health Service Delivery Models (Value-Based)

Module 4: Health System Sustainability & Managing Risk (Week 4)

1. Funding & Payment Models
2. Proactive Risk Management (From Prediction to Monitoring)
3. Resilient Health Systems
4. Aligning Sustainable Development Goals (SDGs)

Module 5: Digital Infrastructure & AI Strategy (Week 5)

1. The Data Lifecycle (Collection and security to analysis and presentation)
2. Dual Perspectives on Artificial Intelligence (Patient versus System Lens)
3. Digital & Algorithmic Governance (Transparency-Ethics-Accountability)

Module 6: Implementation Leadership & Stakeholder Strategy (Week 6)

1. Human Capital Dynamics (Workforce Optimization)
2. Collaborative Leadership (Interprofessional, Multi-Institutional)
3. Stakeholder Mapping (From Identification to Relationship Monitoring)

VII Weekly Schedule & Deliverables

Week	Date	Module	Activities	Deliverables	Weight
1	Jan 13	Found. in Value & Quality	Lecture (90 min)	-	-
			R-Lab (45 min)	Introduction	-
			Group Work (45 min)	Group Formation	-
2	Jan 20	Measuring Value & Quality	Quiz (15 min)	Quiz 1	**5%**
			Lecture (75 min)	-	-
			R-Lab (45 min)	Data	-
			Group Work (45 min)	DAEDALUS	-
3	Jan 27	Health & Delivery	Quiz (15 min)	Quiz 2	**5%**
			Assessment (0 min)	Case Analysis	**30%**
			Lecture (75 min)	-	-
			R-Lab (45 min)	Data	-
			Group Work (45 min)	DAEDALUS	-
4	Feb 03	Sustainability & Risk	Quiz (15 min)	Quiz 3	**5%**
			Assessment (0 min)	R-Visual	**15%**
			Lecture (75 min)	-	-
			R-Lab (45 min)	DAEDALUS	-
			Group Work (45 min)	DAEDALUS	-
5	Feb 10	Digital Infrastructure & AI	Lecture (90 min)	-	-
			Assessment (0 min)	Capstone Brief	**20%**
			Assessment (90 min)	Cap. Pitch (A)	**20%**
6	Feb 17	Leadership & Stakeholder	Lecture (90 min)	-	-
			Assessment (90 min)	Cap. Pitch (B)	**20%**

VIII Weekly Reading List (** Mandatory and * Optional)

Week 1

1. ** Porter ME. What is value in health care? N Engl J Med. 2010;363(26):2477-81.
2. ** Marchildon GP. Health systems in transition. Canada: health system review 2013. 2nd ed. University of Toronto Press; 2013.
3. * Armstrong P, Armstrong H. Decentralised health care in Canada. BMJ. 1999;318(7192):1201-4.

Week 2

1. ** Chiarot CB, et al. Evaluating the concept of access as a critical dimension of universal health coverage in the context of rural communities of Western Province, Zambia. SSM - Health Systems. 2025;5:100106.
2. ** Penchansky R, Thomas JW. The concept of access: definition and relationship to consumer satisfaction. Med Care. 1981;19(2):127-40.
3. ** Donabedian A. The quality of care. How can it be assessed? JAMA. 1988;260(12):1743-8.

Week 3

1. ** Braveman P. Health disparities and health equity: concepts and measurement. Annu Rev Public Health. 2006;27:167-94.
2. ** Porter ME, Lee TH. The Strategy That Will Fix Health Care. Harvard Business Review. 2013.
3. * Sullivan-Taylor P, et al. Integrated People-Centred Care in Canada - Policies, Standards, and Implementation Tools to Improve Outcomes. Int J Integr Care. 2022;22(1):8.

Week 4

1. ** Haw DJ, et al. Optimizing social and economic activity while containing SARS-CoV-2 transmission using DAEDALUS. Nat Comput Sci. 2022;2(4):223-33.
2. ** Haldane V, et al. Health systems resilience in managing the COVID-19 pandemic: lessons from 28 countries. Nat Med. 2021;27(6):964-80.
3. * Kruk ME, et al. What is a resilient health system? Lessons from Ebola. Lancet. 2015;385(9980):1910-2.

Week 5

1. ** Rajkomar A, et al. Scalable and accurate deep learning with electronic health records. NPJ Digit Med. 2018;1:18.
2. ** Floridi L, Cowls J. A Unified Framework of Five Principles for AI in Society. Harvard Data Science Review. 2019.
3. * He J, et al. The practical implementation of artificial intelligence technologies in medicine. Nat Med. 2019;25(1):30–6.

Week 6

1. ** Mitchell RK, et al. Toward a Theory of Stakeholder Identification and Salience: Defining the Principle of Who and What Really Counts. Acad Manag Rev. 1997;22(4):853-86.
2. ** Sutherland J, Hellsten E. Integrated Funding: Connecting the Silos for the Healthcare We Need. Social Science Research Network; 2017.
3. * None.

IX Assessment Portfolio (100%)

Component	Weight	Week Due	Description
Quizzes (Individual)	**15%**	2, 3, 4	15-minute start-of-class quizzes (5% each) based on readings.
Case Analysis (Individual)	**30%**	3	A professional 1,000–1,500 word report on value and quality in a health system.
R Visualizations (Individual)	**15%**	4	A professional technical brief of two (2) figures plus a 200-word insight per figure (400 words total).
Capstone 1A Written Report (Group)	**20%**	5	A professional 10-page (i.e., 3,000–3,500 words) report for the Capstone project.
Capstone 1B Summit Pitches (Group)	**20%**	5 or 6	A 10-minute summit pitch plus 5-minute Q&A for the Capstone project.

X Assessment Objectives & Expectations

Quizzes (15%)

The primary objective of the quizzes is to ensure baseline comprehension of the assigned readings for each week. This prepares students for high-level synthesis during the lecture and active participation throughout the 3-hr session.

1. **Quiz Format:** 15-minute, start-of-class, consisting of ten **(10)** multiple-choice questions.
2. **Platform:** Administered using the MySchulich web portal (i.e., Canvas Learning Management System (LMS); students are **required to bring a laptop** to each session.
3. **Schedule:** Weeks 2, 3, and 4.
4. **Assessment Weight:** Individual-based assessment weighted at 5% of the total course grade (15% total).
5. **Content Scope:** Questions are based strictly on the **assigned readings** for the corresponding week (refer to section VIII Weekly Reading List).
6. **Purpose:** Assessment encourages accountability for preparatory work, ensuring that synchronous lecture time is focused on application and meaningful discussion.

Case Analysis (30%)

Students must select **two (2)** organizations (e.g., companies, health service providers, or technological platforms) and conduct a comparative critique of their performance through the lens of Value and Quality.

Strategic Pillars (Choose One)

The comparison must be framed around **one (1)** of the following module themes:

1. **Effectiveness vs. Efficiency:** *How do the entities balance clinical outcomes with resource optimization?*
2. **Centralized vs. Decentralized Governance:** *How does the organizational structure impact the delivery of high-quality care?*
3. **Public vs. Private Sector Interface:** *How do different funding or delivery mandates influence value for the end-user?*

Evidence-Based Requirements

1. **Empirical Support:** The critique must include at least **one (1)** figure or table (referenced or self-generated) that quantifies a specific value or quality metric (e.g., utilization rates, wait times, or cost-per-outcome).
2. **Analytical Depth:** Students must tie their findings back to the Value Framework:

$$Value = \frac{\text{Outcomes}}{\text{Costs}}$$

3. **Length & Format:** 1,000–1,500 words (**excluding references and appendices**).
4. **Citations:** A comprehensive reference list using the **Vancouver** citation style is required to support the evidence-based arguments. Refer to York University Libaries
5. **Professionalism:** Submissions must adhere to professional executive reporting standards. This includes using a clean, sans-serif font (e.g., Arial or Calibri), an 11- or 12-point font size, and standard 1-inch margins. The report should be logically organized, with clear, descriptive headings and subheadings to guide the reader through the comparative analysis.

R-Based Visualizations (15%)

The objective of this individual assignment is to demonstrate proficiency in utilizing `R` as a strategic instrument to visualize complex health data. Students are required to convert raw data into professional-quality figures that effectively communicate essential insights concerning health system performance.

1. **Deliverable:** Submission of **two (2)** distinct figures generated in `R/RStudio`, each accompanied by a 200-word structured analysis (**400 words total**).
2. **Skill Development & Support:** This assessment is built upon techniques mastered during the weekly R-Laboratory sessions (i.e., Weeks 1–3). Students will utilize a provided health dataset to ensure focus remains on strategic analysis and visualization.

3. **Structured Written Insight:** To ensure clarity and grading equity, each analysis **must** address the following **four (4)** prompts:

   a. **Methodology:** Provide a concise overview of the data processing procedures employed in `R` to prepare the visual, such as filtering, mutating, or pivoting (but not limited to).

   b. **Rationale:** Provide a justification for selecting the specific figure type (e.g., why a boxplot is preferred over a bar chart?) in relation to the particular Value or Quality variable depicted.

   c. **Insights:** Elaborate on the principal findings or patterns disclosed by the visual, explicitly pinpointing deficiencies in effectiveness, efficiency, or equity.

   d. **Recommendation:** A single evidence-based strategic recommendations to a health system leader grounded in the visualized data (minimum of **two (2)** recommendations).

4. **Professionalism & Reproducibility:** Submissions must follow executive reporting standards (standardized fonts/margins) and include the final `R script` to verify reproducibility.

Capstone Written Brief & Summit Pitch (40%)

The Capstone is the *integrative centrepiece of the course*. As designated members of the Health System Taskforce, you and your group will navigate a simulated global health crisis using the DAEDALUS modelling interface.

Access the DAEDALUS Modeling Interface to begin the scenario setup.

Capstone Overview

Objective: To design and defend a nationwide public health response strategy across Canada that maximizes the value and quality of the health system while preserving economic and social resilience.

Simulation: A **new epidemic** has emerged, exerting significant downward pressure on global health systems. The Taskforce must utilize DAEDALUS to model the trade-offs between public health interventions (e.g., sector closures, policy, geography, capacity surges, vaccination coverage, etc.) and macroeconomic stability.

Integrated Support: In Week 2, students will be introduced to the DAEDALUS interface. In Weeks 3 and 4, students will have opportunities for experimentation and data extraction. During the Week 4 R-Laboratory session, standardized code will be provided to assist students in visualizing their specific DAEDALUS simulation outputs in `R/RStudio`.

Capstone Components

Component 1A - The Written Brief (20%): A professional **10-page executive report** (3,000–3,500 words) outlining the strategic response provided by the Taskforce. The report must include a *dual-lens* analysis of public health mandates versus private-sector impacts, supported by at least **two (2)** R-generated visuals derived from the simulation data.

Component 1B - The Summit Pitch (20%): A high-stakes, **10-minute** executive presentation followed by a **5-minute** question-and-answer (Q&A) session. Groups (Taskforce) will deliver their presentation to a *National Cabinet* (i.e., Instructor and Peers), substantiating their strategy's influence on long-term system quality and fiscal value.

Capstone Stakeholder Communication Standards

In the modern healthcare landscape, Value and Quality extend to the clarity and integrity of professional communication. As this course prepares students for executive-level leadership, all deliverables **must adhere** to the following standards of evidence-based communication:

1. **The Executive Briefing Standard** stipulates that all written reports, including the Case Analysis and Capstone Brief, should be regarded as professional briefing reports. Students are expected to: (a) emphasize the *Bottom Line* by presenting the most critical insights (i.e., problem(s)) and strategic recommendations (i.e., solution(s)) at the outset; (b) uphold *Structural Clarity* by employing descriptive headings and sub-headings to facilitate quick comprehension by time-pressed decision-makers; and (c) adopt a *Neutral, Authoritative Tone* to ensure arguments are founded on objective value and quality metrics rather than anecdotal observations.

2. **Data Visualization for Leadership** exemplifies that figures and tables produced with `R code` are not merely technical artifacts; rather, they serve as strategic communication tools. The standards for effective visualization encompass: (a) *Accessibility*, which requires visuals to be interpretable by non-technical stakeholders such as CEOs, ministers, or patients without compromising empirical depth; (b) The *'So What?'* principle, mandating that all visualizations be accompanied by a succinct synthesis elucidating their strategic implications; and (c) *Integrity and Transparency*, ensuring all figures are reproducible, properly labeled, and supported by documented methodologies to maintain stakeholder trust.

3. **Oral Advocacy and The Summit Pitch** are exercises in executive presence and persuasive advocacy. The Taskforce is evaluated on their ability to: (a) synthesize complexity by distilling the multi-dimensional outputs of the DAEDALUS simulation into a **10-minute** cohesive strategy; (b) defend strategic logic by demonstrating analytical rigour during the **5-minute** Q&A session through substantiating claims with references to health system resilience, quality, and value; and (c) engage the *National Cabinet* by addressing the specific priorities of both public-sector regulators and private-sector innovators.

XI Assessment Rubrics

Case Analysis Rubric (30%)

Assessment Criteria	Exceptional (A+)	Proficient (A/B)	Developing (C or below)	Points (100)
Strategic Pillar Analysis	Deeply explores one of the 3 themes (e.g., Efficiency vs. Effectiveness) with nuanced understanding of trade-offs.	Accurately applies a theme to both organizations but may lack depth in exploring complexities.	Fails to clearly link the comparison to a specific module theme or pillar.	**30**
Evidence & Value Framework	Integration of the Value Framework $Value = \frac{\text{Outcomes}}{\text{Costs}}$ is central to the argument.	Mentions the Value Framework but does not fully integrate it into the critique of both entities.	Uses anecdotal evidence; does not apply the formal Value Framework.	**30**
Data Visualization & Support	Includes a high-quality table/figure that is perfectly captioned and directly supports the core argument.	Includes a relevant figure or table, but the connection to the text could be stronger.	Missing a figure/table or uses irrelevant data that does not quantify value.	**20**
Professionalism & Style	Adheres to all executive standards: Vancouver citations, clean sans-serif font, and logical heading structure.	Mostly adheres to standards; minor errors in citation format or formatting consistency.	Significant formatting issues; non-standard font; missing or incorrect citation style.	**20**

R-Based Visualizations Rubric (15%)

Assessment Criteria	Exceptional (A+)	Proficient (A/B)	Developing (C or below)	Points (100)
Technical Execution & Reproducibility	The 'R script' is clean, commented, and runs without errors; figures are exported in high resolution (.png, .jpeg, or .pdf).	Script runs but may lack organization or comments; minor issues with figure resolution or export.	Script contains errors or is missing; figures are poor quality or non-reproducible.	**25**
Data Integrity & Design	Figures use appropriate scales and professional palettes; disaggregation (e.g., by wealth or region) reveals clear equity insights.	Figures are accurate but design choices (colors / labels) could be clearer; disaggregation is present but basic.	Data is misrepresented; missing critical labels; no attempt at disaggregating data for deeper insight.	**25**
Structured Written Insight	All 4 prompts (Methodology, Rationale, Insights, Recommendation) are addressed with high precision and clarity.	All 4 prompts are addressed, but the connection between the *Rationale* and the *Insight* is weak.	Missing one or more of the 4 prompts; analysis is purely descriptive rather than strategic.	**40**
Professionalism	Adheres to executive reporting standards (sans-serif fonts, headings) and stays within the 400-word limit.	Minor deviations from formatting standards or slightly over/under the word count.	Significant formatting errors; ignores word count constraints or executive style.	**10**

Capstone Written Brief Rubric (20%)

Assessment Criteria	Exceptional (A+)	Proficient (A/B)	Developing (C or below)	Points (100)
Simulation Analysis (DAEDALUS)	Expertly uses DAEDALUS data to justify interventions; R-visuals are high-fidelity and insightful.	Correctly interprets model outputs; R-visuals support the narrative but may lack deep strategic linkage.	Model data is misinterpreted or absent; visuals are poor quality or fail to connect to the strategy.	**40**
Dual-Lens Integration	Seamlessly balances public health mandates with private-sector economic resilience.	Addresses both public and private sectors, but the trade-offs between them are not fully explored.	Focuses almost exclusively on one lens (e.g., only health) while ignoring economic/private impacts.	**30**
Strategic Depth & SDG Alignment	Recommendations are proactive, resilient, and clearly aligned with long-term sustainability and SDGs.	Recommendations are logical but reactive; alignment with SDGs is mentioned but not integrated.	Recommendations are vague or unsustainable; no clear alignment with course frameworks.	**20**
Professionalism	Adheres to all executive standards (Vancouver style, formatting, 3,000–3,500 words).	Minor formatting or citation errors; word count is slightly outside the designated range.	Significant lack of professional polish; incorrect citation style; disorganized structure.	**10**

Capstone Summit Pitch Rubric (20%)

Assessment Criteria	Exceptional (A+)	Proficient (A/B)	Developing (C or below)	Points (100)
Narrative & Persuasion	Compelling 10-minute pitch that distills complex trade-offs into a clear, unified strategy.	Clear presentation of facts, but the *strategic story* or persuasive element is less impactful.	Disorganized presentation; exceeds time limit; fails to present a cohesive taskforce strategy.	**40**
Q&A Performance	Defends decisions with *analytical rigour*; substantiates claims using DAEDALUS evidence and course theory.	Answers questions accurately but may struggle to link responses back to empirical data.	Unable to defend strategy; relies on anecdotal answers rather than evidence-based logic.	**40**
Executive Presence	Professional delivery, high-quality slides, and effective taskforce coordination (**all members contribute**).	Professional appearance, but slides may be cluttered or group coordination feels disjointed.	Unprofessional delivery; slides are unreadable; poor group dynamics during the pitch.	**20**

XII R/RStudio Instructions

Overview of R & RStudio

`R` is a powerful, **open-source** programming language specifically designed for statistical computing and data visualization. In this course, we use `RStudio`, an Integrated Development Environment (IDE) that provides a **user-friendly interface** for writing and executing `R code`. This toolset is the **industry standard** for driving evidence-based insights in high-performance health systems.

Getting Started: Download and Install

To ensure compatibility with the R-Lab sessions, students must install both `R` and `RStudio` before the first lecture (i.e., Module 1):

1. **Install `R`:** Visit the CRAN website and download the version appropriate for your operating system (Windows or Mac).
2. **Install `RStudio`:** Visit the Posit website and download the free Desktop version.
3. **Verification:** Open `RStudio`. If you see a version number in the `Console` tab, your installation was successful.

Professional Workflow: `R` Projects and `.Rmd`

We utilize **`R Projects`** (`.Rproj`) to ensure complete reproducibility. An `R Project` locks your working directory to the project folder, eliminating the need for fragile, absolute file paths (like `setwd()`). Within this ecosystem, we write analyses using **`R Markdown`** (`.Rmd`) to seamlessly weave narrative text, mathematical formulas, and executable code chunks into a single executive report.

Setting Up Your Lab Environment:

1. **Create the Project:** Open `RStudio` and navigate to `File > New Project...`. Select **New Directory**, then **New Project**.
2. **Name and Locate:** In the *Directory name* box, type your course identifier (e.g., `MHIA6160_Lab`). Click **Browse...** to choose a permanent, secure folder on your local machine (avoid cloud-synced folders like OneDrive or iCloud during active execution, as they may lock data mid-analysis). Click **Create Project**.
3. **Initialize the Document:** Go to `File > New File > R Markdown...`. A dialog box will appear. Leave the default settings as they are, click **OK**, and a template document will open.
4. **Lock it In:** Immediately go to `File > Save` (or press `Cmd/Ctrl + S`), name the file `lesson_01.Rmd`, and hit save. You will see this file appear in your bottom-right **Files** pane, perfectly paired with your `.Rproj` file.

Installing & Loading Libraries

`R` expands its capabilities through user-contributed packages (`libraries`). Think of installing a package like downloading an app to your phone, and loading a library like opening that app to use it. You only need to install a package **once** on your machine, but you must load it with the `library()` command **every single time** you restart RStudio.

Phase 1: Installing a Package via the `RStudio` Graphic User Interface (GUI) Instead of typing installation code, you can use `RStudio's` built-in interface to download packages safely:

1. Look at the bottom-right pane of your `RStudio` window and click on the **Packages** tab.
2. Click the **Install** button located just beneath the tab bar.
3. In the pop-up window, ensure the *Install from:* dropdown says **Repository (CRAN)**.
4. In the *Packages* text box, type the name of the library you need (e.g., `tidyverse`, `gtsummary`, or `gt`). As you type, `RStudio` will auto complete the name.
5. Ensure the checkbox for **Install dependencies** is checked.
6. Click **Install**. You will see red text scroll by in your bottom-left Console pane—this means the download is active. Wait until the blue `>` cursor reappears, signaling the installation is complete.

Phase 2: Activating the Package in Your Script A downloaded package sits dormant on your hard drive until you explicitly wake it up. To make the package functions available within your `R Markdown` script, you must type and execute the `library()` command inside a code chunk.

For example, to activate the core data suites we will use in this course, include and run this block at the very top of your document:

```r
# Execute this chunk to activate your packages for this session
library(tidyverse)  # For data manipulation and visualization (includes ggplot2)
library(gtsummary)  # For professional health-related tables
library(gt)         # For rendering professional executive tables
```

Phase 1 & 2 Alternative: Students who prefer an entirely code-driven setup can copy and paste the following block directly into their `R Markdown` (`.Rmd`) file to prepare their environment.

```r
# 1. Install necessary libraries (Run this section only once)
install.packages("tidyverse")   # For data manipulation and visualization
install.packages("gtsummary")   # For professional health-related tables
install.packages("gt")          # For high-quality table formatting

# 2. Load the libraries (Include this block at the top of every script)
library(tidyverse)
library(gtsummary)
library(gt)

# 3. Verify setup
print("R Environment is ready for Value and Quality analysis.")
```

Pro Tips for Success:

1. **The Help Command:** If you are unsure how a function or package works, type a question mark before the command in your console to bring up the official documentation in the `Help` pane:

```r
?tidyverse
?gtsummary
?gt
```

2. **Data Reproducibility:** Always create a sub-folder named data inside your main `R Project` directory. Keep all raw files (`.csv`, `.xlsx`) there. This allows you to write clean, portable import paths like `read_csv("data/health_utilization.csv")` that will run effortlessly on any machine.

XIII Final Course Grade & Policy

Course assessments will generally be evaluated based on a percentage scale. Each assessment carries a specific weight as specified in the course outline. **Final grades are calculated by multiplying each assessment's percentage score by its designated weight, summing these values to derive a total percentage out of 100, and then converting this total percentage into a letter grade as specified below**. Within this framework, the course will **not** employ a grade point scale. On occasion, higher letter grades may be awarded; however, the grades assigned will never be lower than those determined by the standard scale. The grading scheme for master's level courses at the Schulich School of Business employs a **nine-point grade-point** system as follows:

Letter Grade	Stanine	Percentage
A+	9	95+
A	8	85–94
A-	7	80–84
B+	6	77–79
B	5	74–76
B-	4	70–73
C+	3	67–69
C	2	64–66
C-	1	60–63
F	0	–

Grades at the Schulich School of Business are based on a **9-point grading scale**. The highest grade attainable is A+ (9), while the minimum passing grade is C- (1). **To ensure comparability of final grades across different courses, elective courses are expected to have a mean grade ranging between B (5.2) and B+ (6.2)**. The Schulich School of Business **does not** implement a percentage scale, **nor** does it prescribe a standard conversion formula from percentages to letter grades.

Adjustments of grade conversions within individual courses are at the discretion of the instructor. For further details regarding the grading index, grading policies, and grade point average (GPA) requirements, please refer to your MHIA Student Handbook.

XIV University Guidelines & Policies

Academic Honesty

Academic integrity is a **core value** of the Schulich School of Business and York University. Students are expected to maintain the highest standards of honesty.

1. **Originality:** All individual assignments (Case Analysis, R-Visualizations) must be the student's own original work.

2. **Collaboration:** While the Capstone project is a group effort, *unauthorized collaboration* on individual assignments is **strictly prohibited**.

3. **Citations:** Proper attribution is required for all sources using the **Vancouver** citation style. Failure to cite sources constitutes plagiarism.

Accommodations for Students with Disabilities

York University is committed to providing an accessible learning environment.

1. **Support:** Students with physical, sensory, or learning disabilities who require accommodations are encouraged to contact Student Accessibility Services early in the term.

2. **Notification:** Please provide the instructor with the official accommodation letter as soon as possible to ensure that your learning needs are met, particularly for the timed quizzes.

Missed Quizzes and Deadlines for Assessments

Given the condensed 6-week nature of this course, adherence to deadlines is critical.

1. **Quizzes:** There are **no** make-up quizzes. Since the quizzes are administered at the start of class, punctuality is essential.There will be **no** extra time given to complete the quizzes.

2. **Late Assessments:** Assessments submitted after the deadline will be subject to a penalty (**10% per day**) unless prior arrangements have been made due to documented medical needs.

Classroom Etiquette & Professionalism

In alignment with our Stakeholder Communication Standards (refer to above), students are expected to behave as professionals.

1. **Technology Use:** Laptops are required for R-Lab sessions and quizzes. However, non-course-related use of technology during lectures is discouraged to maintain a high-quality learning environment.

2. **Engagement:** Active participation in Group Work and the Summit Pitch is expected from all members to ensure Collaborative Leadership.

Grading Scheme

Grades at the Schulich School of Business are awarded on a **letter-grade basis** (e.g., A+, A, B). All grades are reviewed and approved by the Area Coordinator and the Associate Dean to ensure grading equity across the program.

Note: For a complete list of policies, please refer to the student handbook.

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